

Mega3 + Mega4 Simulation Analysis Report

Deep statistical analysis of the `results/mega3/` sweep (first mega run with the T.30 ability catalog wired in) plus the `results/mega4/` re-run (corrected `expected_wr` model + $5\times$ sample). Successor to `reviews/mega2_analysis_report.md`.

Generated 2026-06-01, mega4 integrated 2026-06-02, via R 4.5.3. All figures, tables, and scripts are reproducible — see §14.

Read §0.5 first. Mega4 fixed a bug in `expected_wr`. Combat is unchanged, so **every win_rate finding below holds** (confirmed at $5\times$ sample). But **every `wr_delta` figure carried in §2, §4, §6, §9, §11 was computed on the buggy mega3 metric** — the corrected (mega4) values are in §0.5, §4, and §9. The buggy values can even change sign (role `wr_delta`), so trust only the explicitly-labelled **corrected** numbers.

0. Executive summary

This is the first dataset where designed ability kits actually fire. Mega2 ran on a build where 0/240 roster ability-ids resolved to a handler, so every piece fell back to an INT-scaled generic nuke. Mega3 inherits the T.30 catalog. The headline is the **consequence of that fix**:

Implementing kits flipped the role hierarchy. The INT-nuke artifact that propped mages up and crippled STR pieces is gone — but the pendulum overshot. **Marksman (+0.17) and warrior (+0.12) recovered exactly as mega2 predicted; mage collapsed (−0.17) to the lowest win rate.** (Caveat that the whole report turns on: **raw role win rate is confounded by tier** — see §4. Once you control for tier / read the corrected `wr_delta`, the genuinely **over-tuned** roles are marksman & warrior, and both **mage and hybrid** **under*-deliver* vs their power target.)

Four findings carry the report:

1. **Aggregate balance is still healthy.** Champion-vs-enemy win rate is centered on 0.50 in every format (§2). The faction math is sound; the problems are **within-roster**.
2. **Mage is broken at every tier** — a ~ 0.20 win-rate deficit vs non-mages **after controlling for tier**, in both champion and enemy factions (§4–5). Not a faction or a content bug — a role-design failure.
3. **The tier cliff survived the kit rewrite.** Parity sits at tier ~ 6 ; tiers 1–5 are all net losers; `cor(tier, win_rate) = 0.87` (§6). **(The original "high tiers beat their power budget" claim rested on a buggy `expected_wr` — corrected in §0.5.)**
4. **Weather is nearly inert.** Own-weather affinity is worth only **+0.011** win rate; the strongest piece posts an **identical** 0.9958 across clear/cloudy/mist/rain. The core thematic mechanic barely touches outcomes (§8).

Plus two sharp piece outliers: **Grand Marshal** (clear warrior T10) is effectively unloseable at 0.9958 in 1v1; **Will-o-Fawn** (mist mage T2) is near-unwinnable at 0.0504.

0.5 Mega4 integration & the `expected_wr` fix

Added 2026-06-02. A second sweep, `results/mega4/`, was run after this report's first draft. It changes **two** things and nothing else:

1. **`expected_wr` bug fixed.** Mega3 derived `expected_wr` from a Bradley-Terry beta model; mega4 replaces it with a **deterministic power-threshold model** (`tools/simulation/ratings.py:85-95`): for each matchup, `expected` = 1.0 if your team's total `power(T,L)` exceeds the opponent's, 0.0 if less, 0.5 if equal — averaged over the **actual** opponent field. So `wr_delta = win_rate - expected_wr` now reads cleanly as **"do you beat the opponents you outpower?"**

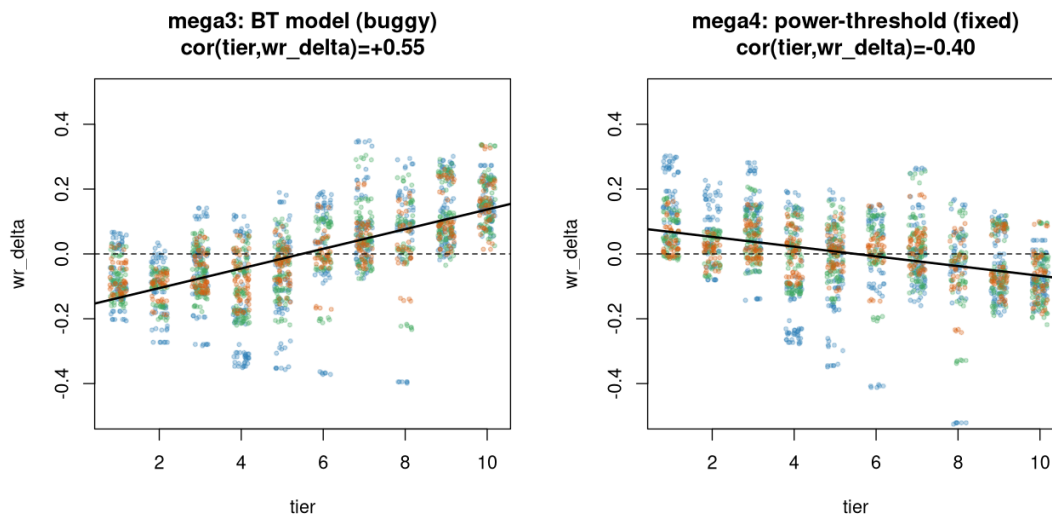
2. **5× sample.** team2 n=130k (was 26k), team3 n=78k (was 16k). 1v1 stays full round-robin.

The combat engine did not change. Per-piece `win_rate` is identical between runs (Aegis Tortoise = 0.4034 in both; median per-piece 1v1 shift = **0.0000**). Therefore:

- [OK] Every `win_rate`-based finding (§2, §3, §4, §5, §7, §8, §9, §10) holds — and is now confirmed at 5× sample. Role means move <0.002 mega3→mega4; the mage within-tier deficit is **0.198** in both; own-weather advantage is +0.008 in both. Mega3 was already converged.
- [X] The `wr_delta` interpretation in the original §6 and §11 was computed on the buggy metric. It is corrected here and those sections are annotated.

[!] Mega4 is **still running** as of 2026-06-02 07:20 — `results/mega4/` has 1v1 ×6 and team2 ×6 complete, but **team3 only has clear/cloudy/mist/rain (4/6 — snow + thunder pending, ETA ~2h)**. All `wr_delta` figures below use complete cells only; team3 numbers will firm up when the run finishes. A stale background mega3 job (small-n) is also alive and contending CPU — worth killing to speed mega4 (see closing note).

Corrected calibration finding (replaces old §6 claim #3):



expected_wr calibration fix

What `wr_delta` actually measures. Power P is a **fixed design target** assigned to each piece — it does not move. Pieces are tuned (active/passive abilities + stat fine-tuning) so that each one's **realized** combat strength matches its assigned P . The corrected `expected_wr` is the win rate a piece **should** post if it exactly hit its P target against the field it faced. Therefore:

`wr_delta` is the per-piece tuning residual. `wr_delta` ≈ 0 → the piece is tuned on-target. `wr_delta` < 0 → **under-tuned** (its kit delivers less than its P promises). `wr_delta` > 0 → **over-tuned** (kit over-delivers vs P). The tuning goal is to drive every piece's `wr_delta` toward 0 — **not** to change P .

The buggy BT model produced `cor(tier, wr_delta)` = +0.55→+0.74. Under the fixed power-threshold model the correlation **flips to -0.40** (stable across all three stages) and the spread tightens (`sd(wr_delta)` 0.160→0.147 in 1v1, 0.107→0.074 in 3v3). The corrected reading:

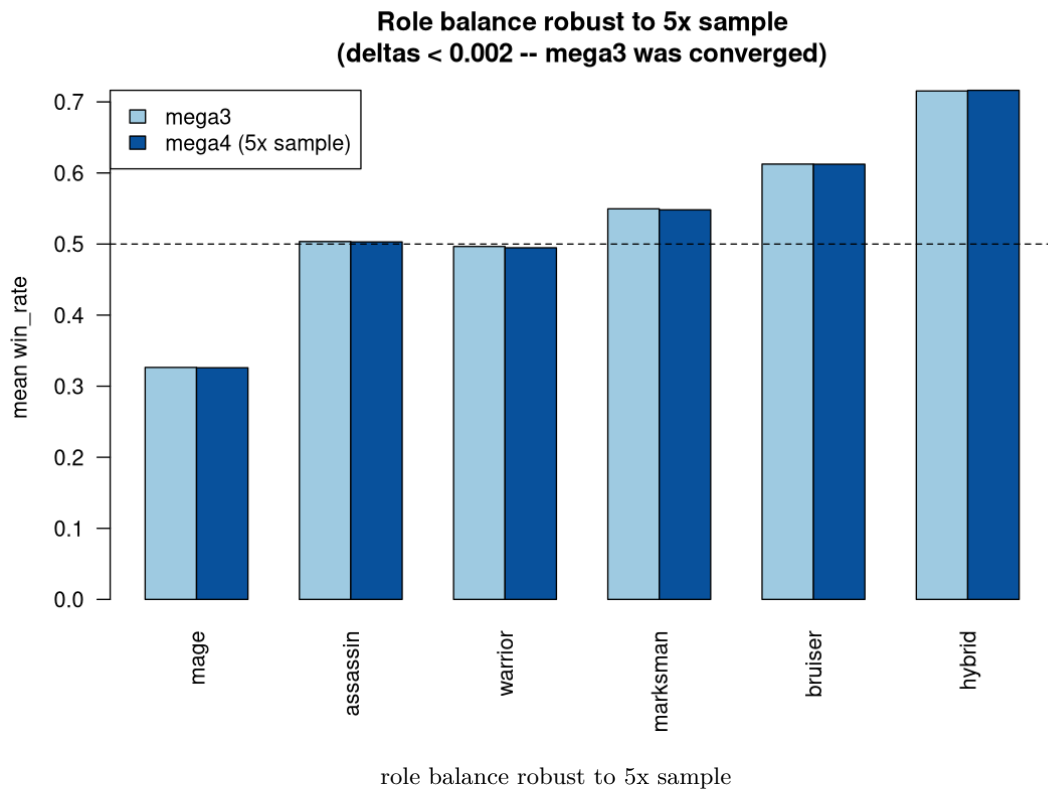
- **There is a mild systematic under-tuning at high tiers.** The negative slope says high- P pieces realize **slightly less** than their target (their bigger kits don't fully cash in) while low- P pieces slightly over-deliver. It's a small residual, not an emergency — but it points tuning effort toward the top of the roster.

- **power() does not need re-fitting.** P is the fixed yardstick by design; the earlier "re-fit power()" idea was backwards. The lever is each piece's abilities/stats, measured *against* P.
- **The tier win_rate cliff is intended, not a bug** ($\text{cor}(\text{tier}, \text{win_rate}) = 0.87$): a high-P piece *should* beat a low-P one. §6's cliff is by-design decisiveness; only its old `wr_delta` gloss was wrong.

Corrected outlier reading (refines §11): with a trustworthy `wr_delta`, the weak-mage list splits into two kinds —

Genuine underperformers (lose vs equal/weaker power)	"Low but on budget" (just low-tier)
Hierarch T8 (−0.379), Marsh Thrush T6 (−0.264), Company Captain T5 (−0.209), Storm Eagle T9 (−0.158), Arcanist T9 (−0.111), Glade Heron T8 (−0.128)	Will-o-Fawn T2 (−0.069), Signal Drummer T1 (+0.009), Sparkfly T1 (+0.019), Dusk Bat T2 (−0.039)

In tuning-target terms: the red column = mages whose kits are **under-tuned vs their assigned P** (they have the target strength to win and don't); the grey = mages that are **on-target** and simply low-P. This **sharpens the mage verdict (§5)**: the tuning work is concentrated in **mid/high-tier mages** (Hierarch T8 −0.38, Marsh Thrush T6 −0.26, Company Captain T5 −0.21, Storm Eagle/Arcanist T9) — buff their abilities until `wr_delta` → 0. The tier-1/2 mages (incl. Will-o-Fawn, the headline "0.05" piece) are mostly on-target — their low win rate is their low P, not a tuning fault. **Mage fix priority = T6+ casters (Hierarch first), not the T1 floor.**





Also note **Grand Marshal**: his `wr_delta` drops from +0.31 (buggy) to +**0.07** (fixed) — the threshold model correctly *expects* a T10 to win ~0.96, so he’s no longer a model-flagged overperformer. He is still a raw outlier (0.97 win, near-unloseable) and still worth a nerf; he’s just winning about what his power says he should.

1. Dataset and method

Stage	Weathers	Battles / weather	Pieces	Rated rows
1v1 (round-robin)	6	7,140	120	720
team2-sample (2v2)	6	26,000	120	720
team3-sample (3v3)	6	16,000	120	720

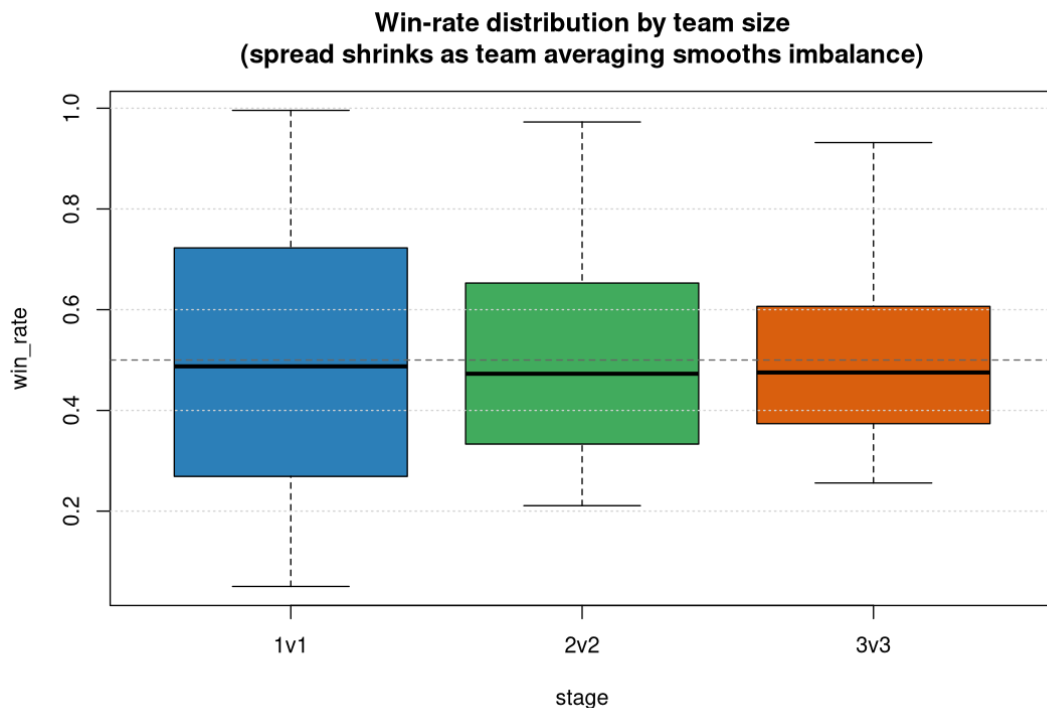
- **294,840 battles** total across 18 `results_*.csv`; **2,160 rated rows** (120 pieces × 3 stages × 6 weathers) across 18 `ratings_*.csv`. (Mega4 re-runs the same design at ~5× sample — team2 130k/wx, team3 78k/wx — see §0.5.)
- 120 pieces = 60 champions + 60 enemies. Affinity split is unbalanced by design: **clear 40, all others 16** (clear is the neutral default).
- `--max-ticks 12000` this run (vs mega2’s effectively-infinite 1e6) → **timeouts now exist** (~11–13% of fights). Every timeout is recorded as a **draw** (verified: `draw == timed_out` to the row in all stages). This is the main reason duration/timeout metrics are *not* comparable to mega2.
- **Metrics:** `win_rate`; `expected_wr`; `wr_delta = win_rate - expected_wr`; `timeout_rate`; `mean_duration_ticks`. `expected_wr` differs by run: mega3 used a Bradley-Terry budget (since shown buggy — it also emitted `beta/beta_ratio/beta_deviation_pct`); **mega4 uses the corrected deterministic power-threshold model** (§0.5). Read `wr_delta` only from mega4.
- **Power proxy** for the win-curve: team ΣP via `power(tier, 1)` mirroring `src/game/scaling.py`. Per-battle level isn’t logged, so this is a level-1 proxy — adds scatter, preserves monotone shape.

2. Aggregate balance is healthy

Champion vs enemy mean win rate, symmetric and centered everywhere:

stage	champion	enemy
1v1	0.516	0.484
2v2	0.508	0.494
3v3	0.500	0.501

The ~1.6pp champion edge in 1v1 dissolves with team size. Faction-level balance is **not** a problem. Every issue below lives **inside** the roster.



win-rate distribution by stage

The distribution **tightens dramatically with team size** — $\text{sd}(\text{win_rate})$ falls 0.267 (1v1) $\rightarrow$ 0.187 (2v2) $\rightarrow$ 0.149 (3v3). (Model-independent, so unaffected by the `expected_wr` fix.) Team averaging launders individual imbalance: a broken piece on a 3-stack is diluted by two teammates. This is structural and matches mega2.

3. The kit-implementation flip (mega2 $\rightarrow$ mega3)

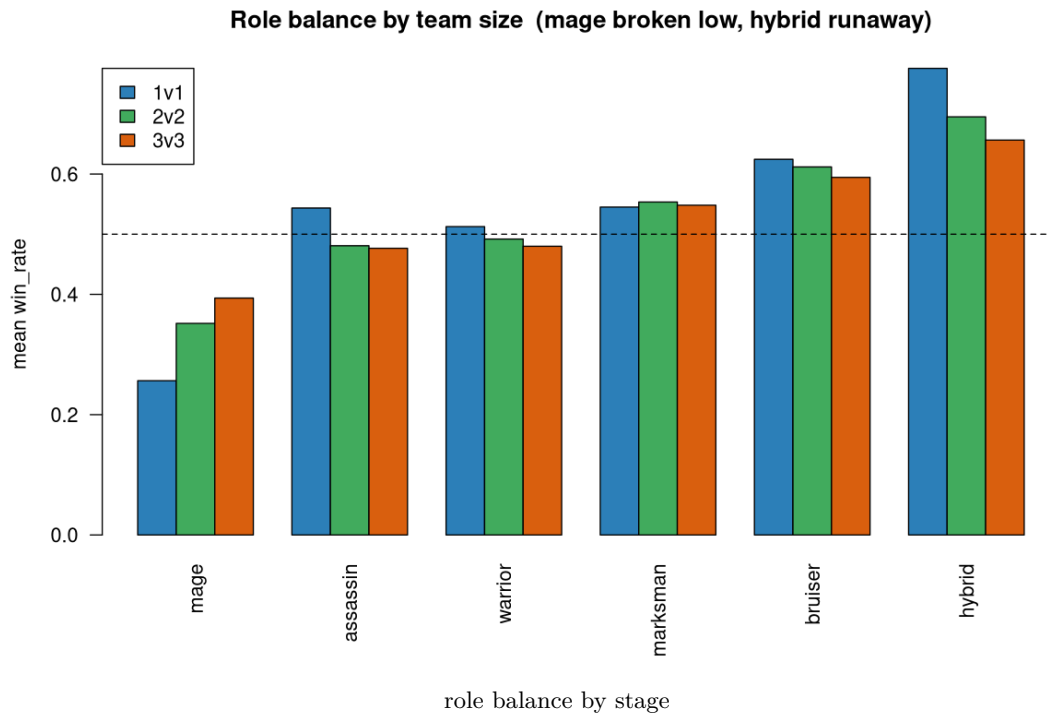
Mega2's rectification note warned that the role verdicts were artifacts of the INT-scaled fallback nuke (`raw = 0.2 \cdot \text{STR} + 4.2 \cdot \text{INT},  $\text{INT} \geq 21 \times \text{STR}$ ). Mega3 confirms the warning and the predicted direction of correction:`



role	mega2 (1v1)	mega3 (1v1)	Δ
marksman	0.374	0.545	+0.171
warrior	0.395	0.513	+0.117
bruiser	0.598	0.625	+0.026
hybrid	0.782	0.776	-0.007
assassin	0.594	0.544	-0.051
mage	0.421	0.256	-0.165

STR/AD pieces (marksman, warrior) — ”crippled by the INT fallback, not mis-designed” per mega2 — **recovered to ~parity** the moment their real kits fired. But mage, which the fallback had been silently carrying, **fell off a cliff** once it had to rely on its actual kit. The fix was correct; the mage kit catalog is the new gap.

4. Role balance — tier-confounded; mage & hybrid under-deliver, marksman/warrior over-deliver



[!] **Read raw role win_rate with care — it is confounded by tier.** Roles are not evenly spread across tiers: mage mean tier **4.1**, hybrid mean tier **9.1** (median 10). So "mage lowest / hybrid highest win_rate" largely restates "mage pieces are low-tier, hybrid pieces are high-tier." The honest role signal is the **corrected wr_delta** (delivery vs the P target, §0.5) and the **within-tier** comparison below.

Overall mean win rate (raw) and **corrected (mega4) wr_delta** by role:

role	win_rate	mean tier	wr_delta (corrected)	reading
mage	0.334	4.1	−0.064	low win = low tier; under-delivers, worst at high tier (§5)
assassin	0.500	5.9	−0.014	on-target
warrior	0.495	4.5	+0.072	over-target (wins more than P warrants)
marksman	0.549	5.0	+0.087	most over-target
bruiser	0.610	6.5	+0.041	mildly over-target
hybrid	0.709	9.1	−0.082	**most *under*-target — high win_rate is pure tier**

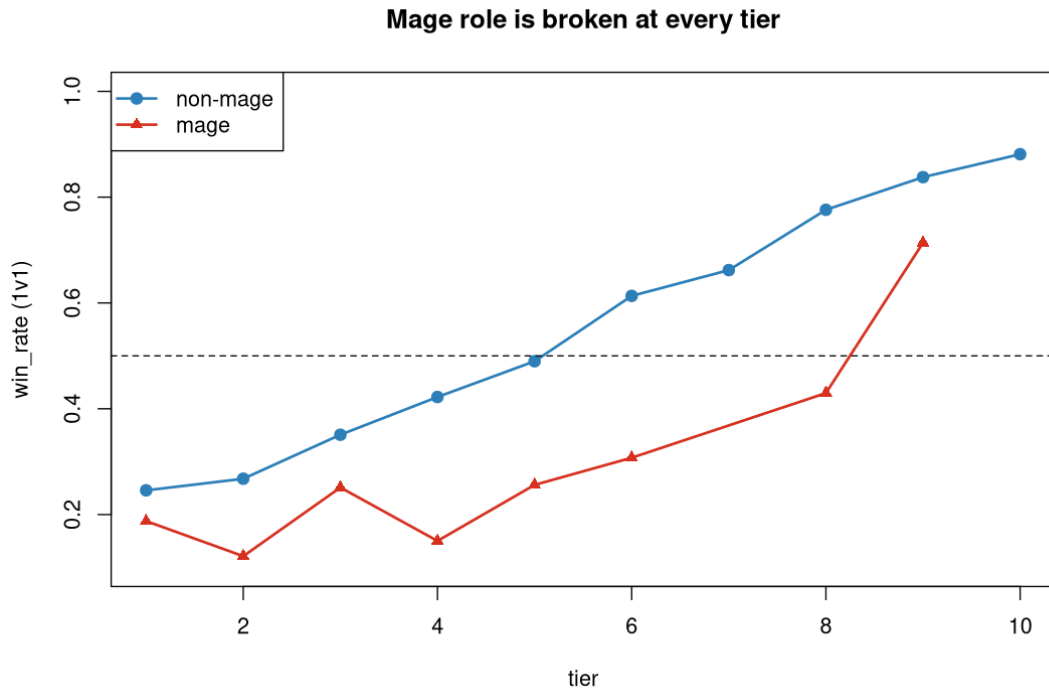
The corrected **wr_delta** ranking is **nearly the reverse** of the raw-win_rate ranking. Two reframes follow:

- ****Hybrid is *not* a runaway over-powered role. Its 0.71 win_rate is a roster-composition artifact: hybrids are almost all tier 9–10.** Within tier, hybrids win 6–9pp *less* than non-hybrids at the same tier** ($\Delta = -0.057$ to -0.091 ; mean -0.082) — they *under*-deliver their P, the same direction as mage. No nerf is warranted; if anything hybrid kits are slightly under-tuned. *(The earlier "hybrid runaway / design debt" read was an un-tier-controlled error.)*

- **The genuinely over-target roles are marksman (+0.087) and warrior (+0.072)** — they win more than their (lower) P says they should. If any role is "too strong for its power," it's these two, not hybrid.

So the under-delivering roles (buff candidates) are **mage and hybrid**; the over-delivering roles (trim candidates) are **marksman and warrior**; assassin/bruiser are close to on-target. Mage stays the priority because its under-delivery concentrates in expensive high-tier casters (§5) and it also owns the raw-win_rate floor.

5. Mage deep-dive — the diagnosis



mage vs non-mage by tier

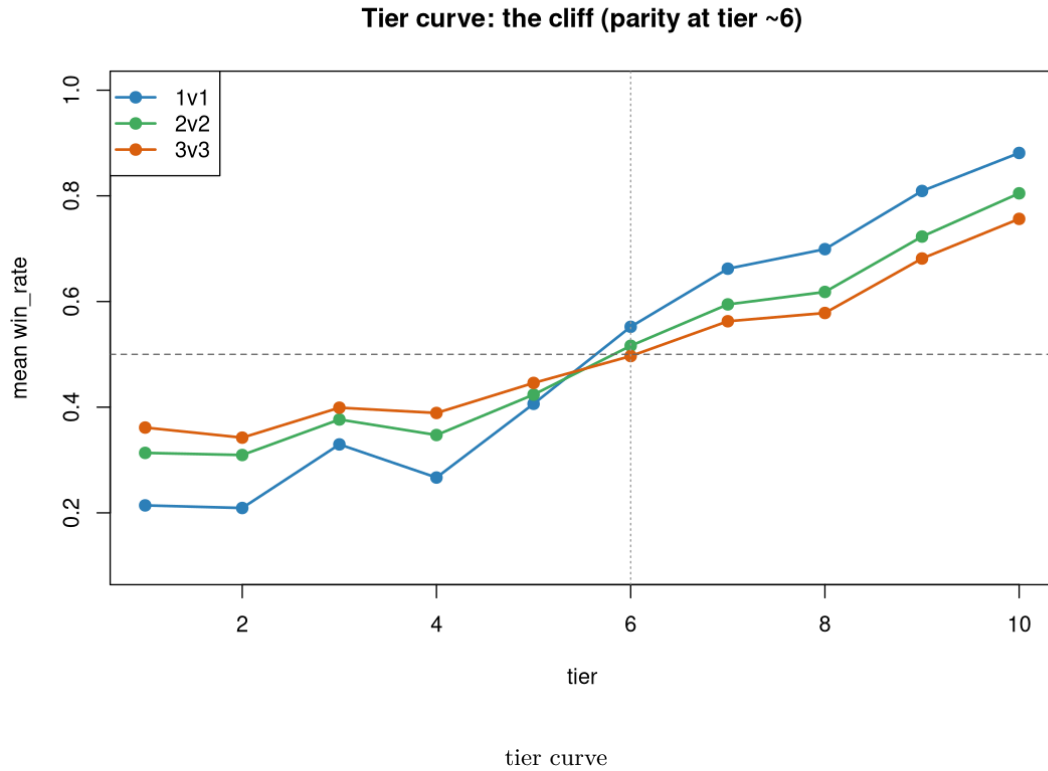
The mage line tracks ~0.15–0.25 *below* non-mages at **every** tier. A tier-9 mage barely reaches parity; the roster has no tier-10 mage to test the ceiling.

Controls that rule out the easy explanations:

- **Not a faction bug.** Mage champions (0.355) and mage enemies (0.309) are *both* broken. It is the role, not one content author.
- **Not a tier artifact.** Mean within-tier deficit vs non-mages (1v1) = **0.198**. Even matched on tier, a mage loses ~20pp it shouldn't.
- **Mechanism = no burst, no bulk.** Mages have the lowest HP and now lack the free INT nuke that the fallback gave everyone. Their fights run *longer* than average (7157 vs 6954 ticks in 1v1) and they still lose — they neither kill fast nor survive. timeout_rate (0.15) is mid-pack: they don't stall to a draw, they get ground out.

This is the report's top action item. The mage kit catalog either deals too little damage, fires too slowly, or both. Re-tune mage abilities (burst windows / cast cadence / base HP) and re-sim before any other balance pass.

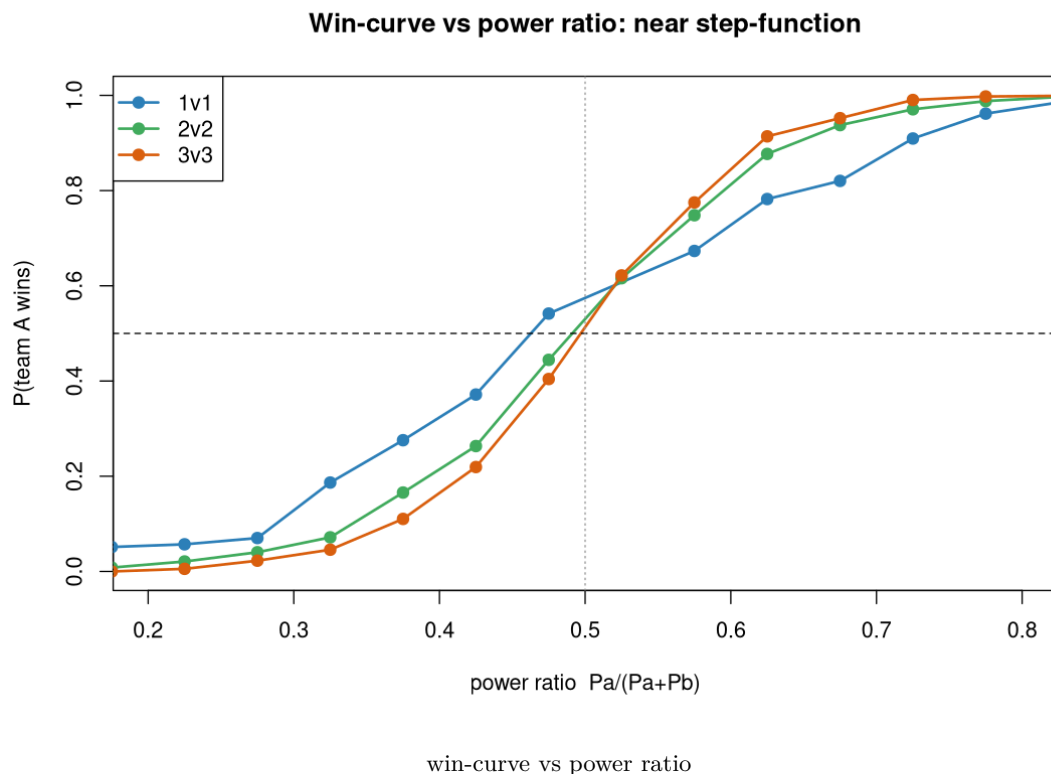
6. The tier cliff



The cliff that dominated mega2 **survived the kit rewrite intact**. $\text{cor}(\text{tier}, \text{win_rate}) \approx 0.87$ in all three stages. Parity crosses at **tier ~6**: tiers 1–5 are net losers, 7–10 net winners. The low-tier floor is brutal — tier-1/2 pieces win ~21% in 1v1. This is an observed-outcome fact, independent of any rating model, and it is **confirmed unchanged at mega4's 5× sample**.

[!] **Corrected** — see §0.5. This section originally argued from `wr_delta` that "high tiers beat their power budget" ($\text{cor}(\text{tier}, \text{wr_delta}) = +0.55 \rightarrow +0.74$) and that `power()` under-rewards low tiers. **That rested on a buggy expected_wr**. Under mega4's fixed power-threshold model the correlation flips to **−0.40** and the spread tightens — there is **no** systematic under-reward of low tiers. The `power()` curve does **not** need re-fitting. The cliff is about HP/DPS/focus-fire decisiveness (§7), not a scaling-math error. The plot below (`06_wrdelta_vs_tier.png`, buggy metric) is retained only for the before/after in §0.5.

7. Decisiveness and the win-curve



Combat is still a **near step-function** of power ratio — the contested band (where $P(\text{win}) \in [0.2, 0.8]$) is roughly $P_a/(P_a+P_b) \in [0.35, 0.65]$. Two refinements over mega2:

- **3v3 is the steepest** (most decisive at parity), 1v1 the flattest. More bodies → more reliable focus-fire → less variance → sharper curve. Consistent with the `sd(win_rate)` shrink in §2.
- **1v1 carries a side-A bias.** The 1v1 curve crosses 0.5 *left* of parity; at $|pr-0.5| < 0.025$, side A wins **0.542** in 1v1 vs ~0.51 in 2v2/3v3. A ~4pp initiative/turn-order advantage in duels. Worth checking the 1v1 tie-break / who-acts-first logic.

Winner HP-remaining (stomp proxy) scales with team size — 799 (1v1) / 1406 (2v2) / 1937 (3v3) — i.e. larger fights end with more leftover HP on the winning side: snowball/focus-fire leaves survivors at high HP.

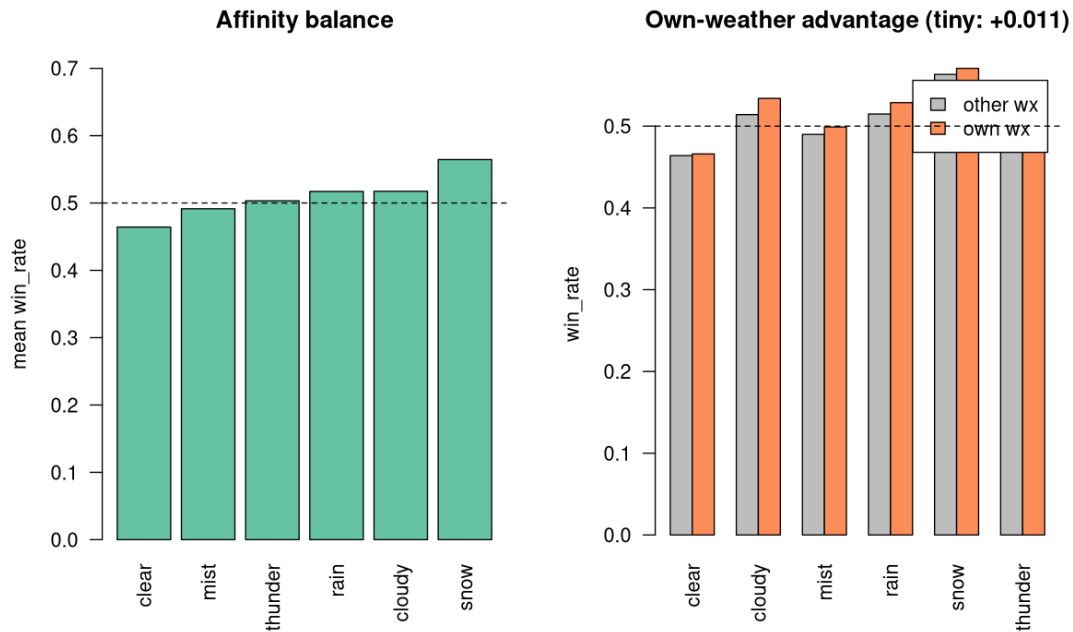
Balance scorecard (fraction of piece×stage×weather cells near parity):

stage	in [.45,.55]	blowout <.3	blowout >.7
1v1	0.11	0.28	0.28
2v2	0.17	0.14	0.18
3v3	0.25	0.04	0.12

1v1 is **highly polarized** — only 11% of matchups are competitive, 56% are blowouts. This is the cliff (§6) plus the role tails (§4) compounding. It tightens at 3v3 but never gets *good*.

8. Weather is nearly inert

The signature mechanic — real-world weather shaping combat — barely registers.



affinity balance and own-weather effect

- **Own-weather advantage = +0.011 win rate.** A piece fighting in its own affinity-weather wins 0.509 vs 0.499 otherwise. Largest sub-effect is thunder (+0.024); clear is ~0.
- **Mean per-piece win-rate range across all 6 weathers = 0.027** (median 0.022). The *most* weather-sensitive piece (Aerion) swings only 0.090.
- **Smoking gun:** Grand Marshal posts an *identical* 0.9958 in clear, cloudy, mist, and rain (§11). Will-o-Fawn posts an identical 0.0504 across mist/rain/snow/thunder. Outcomes are weather-invariant to 4 decimals for the pieces where it would matter most.

Either the weather stat-packs / affinity damage-triangle apply tiny coefficients, or kits don't read weather state. For a game whose pitch is "live weather shapes combat," a $\pm 1\text{pp}$ effect is a thematic failure. **Recommend a dedicated weather-sensitivity audit** before shipping — this is a top-3 issue even though it isn't a *balance* issue.

9. Affinity balance

`wr_delta` shown is the **corrected (mega4)** value:

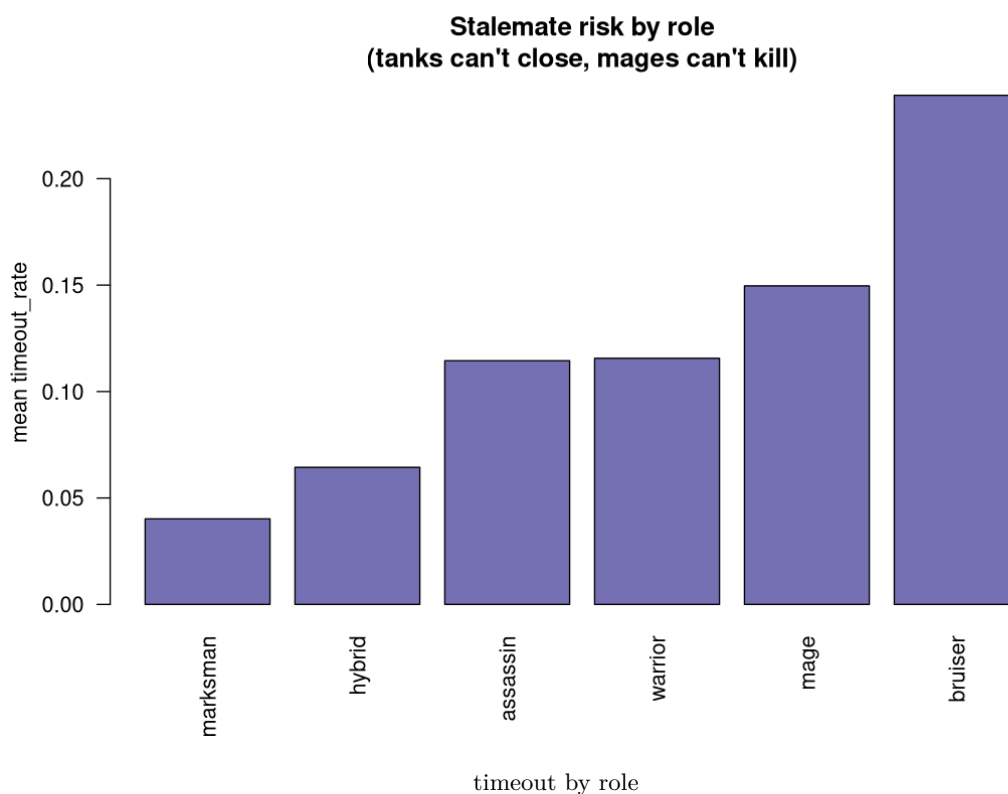
affinity	win_rate	wr_delta (corrected)	n pieces
clear	0.464	+0.010	40
mist	0.491	−0.033	16
thunder	0.503	−0.023	16
rain	0.517	−0.007	16
cloudy	0.517	−0.005	16
snow	0.564	+0.042	16

Spread is modest (win_rate 0.46–0.56). **Snow runs hot** on both metrics (+0.042 over target). Under the corrected model, **clear is actually on-target** (+0.010) — its low win_rate is just its low-tier-heavy roster (40 pieces incl. most low-tier mages), not a tuning fault; **mist/thunder mildly under-deliver** (−0.03/−0.02). All second-order next to mage/tier/weather. *(Note the sign flip vs the buggy mega3 values — clear was −0.016, now +0.010.)*

10. Stalemates and timeouts

With `--max-ticks 12000`, ~11–13% of fights now time out ($\rightarrow$ draw). Distribution by role:

role	timeout_rate
bruiser	0.239
mage	0.150
warrior	0.116
assassin	0.114
hybrid	0.064
marksman	0.040



Two distinct stall modes: **tanks that can't close** (bruiser/warrior — high HP, low burst $\rightarrow$ mutual attrition) and **mages that can't kill** (low damage). Marksman/hybrid resolve cleanly. The worst single piece is **Coral Colossus** (warrior T5, 46% timeouts, mean 10,063 ticks) — a damage-starved wall. Bruiser timeouts are the strongest argument for keeping a finite max-ticks + sudden-death rule rather than mega2's 1e6.

11. Outlier roster

Strongest (mean win_rate, all stages/weathers):

name	affinity	role	tier	win_rate	wr_delta
Grand Marshal	clear	warrior	10	0.963	+0.311
Storm Tyrant	thunder	hybrid	10	0.882	+0.228
Aurion, the First Dawn	clear	hybrid	10	0.882	+0.228
Sunspear Falcon	clear	marksman	9	0.873	+0.258
Quarried Behemoth	cloudy	bruiser	9	0.864	+0.248
Glacierback Mammoth	snow	bruiser	7	0.828	+0.282

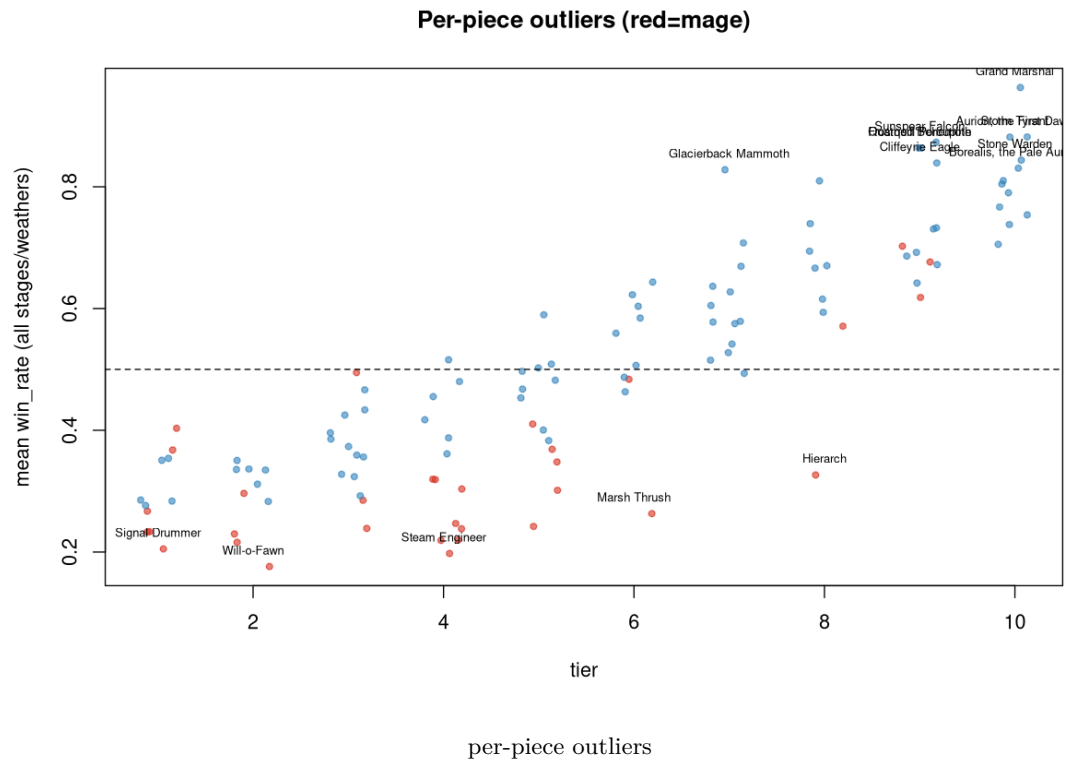
Weakest — every one is a mage:

name	affinity	role	tier	win_rate	wr_delta
Will-o-Fawn	mist	mage	2	0.176	−0.218
Steam Engineer	clear	mage	4	0.198	−0.247
Signal Drummer	clear	mage	1	0.205	−0.162
Dusk Bat	cloudy	mage	2	0.216	−0.179
Coppercrest Stork	thunder	mage	4	0.219	−0.228
Drowned Siren	rain	mage	4	0.220	−0.230

[!] The `wr_delta` column in both tables above uses the buggy `mega3` metric. For the trustworthy tuning-residual ranking, use the corrected values in §0.5. `win_rate` (raw outcome) is correct as shown.

Specific flags (with corrected `mega4 wr_delta`):

- **Hierarch** (clear mage T8) — corrected `wr_delta` = **−0.379**, the single most under-tuned piece in the roster: a high-P caster whose kit delivers nowhere near its target. **Top buff target.**
- **Grand Marshal** — hits **0.9685 overall / 0.9958 in 1v1, weather-invariant**, near-unloseable. But corrected `wr_delta` is only **+0.07** — he’s winning roughly what his T10 P *says* he should. He’s a “feels-bad invincible” UX problem more than a tuning-error problem; nerf only if his P target itself is meant to be lower.
- **Marsh Thrush** (T6 mage, −0.264) / **Company Captain** (T5 mage, −0.209) — next under-tuned casters after Hierarch.
- **Glacierback Mammoth** (T7 bruiser) — corrected `wr_delta` $\approx$ +0.08, mildly over-target; minor trim at most (the buggy +0.282 over-stated it).
- **Will-o-Fawn** — 0.168 win, weather-invariant, but corrected `wr_delta` $\approx$ −0.07: **on-target for its T2 P**. Not a tuning fault; previously mis-flagged as “bottom of the crisis.”



12. Prioritized recommendations

All recommendations are framed as **tuning pieces toward $wr_delta \rightarrow 0$ against their fixed P targets** — never as changing P.

#	Action	Why	Effort
1	Buff the mage kit catalog, T6+ first (Hierarchy, Marsh Thrush, Company Captain, Storm Eagle, Arcanist)	Mage is the one broken role (0.33, −0.20 within-tier, both factions, §5); the corrected residual localises the fault to mid/high-tier casters that under-deliver vs P (§0.5)	High
2	Audit weather coefficients / confirm kits read weather	Own-weather worth +0.008–0.011; outcomes weather-invariant to 4dp — the core mechanic is inert (§8)	Med
3	Drive wr_delta → 0 roster-wide; close the −0.40 high-tier under-tuning slope	wr_delta is the tuning-error signal vs the fixed P target; do not re-fit power() (§0.5). High-P kits systematically under-deliver	Med
4	Buff Hierarchy (−0.38, most under-tuned piece)	Largest single tuning residual in the roster (§11)	Low
5	Trim marksman & warrior, or buff hybrid — they're the real per-P outliers	Corrected wr_delta: marksman +0.087 / warrior +0.072 over -deliver; hybrid −0.082 under -delivers (its 0.71 win is just high tier). Hybrid is <i>*not*</i> over-powered (§4)	Med
6	**Decide if Grand Marshal's *target* is correct**	0.97 win / near-unloseable, but wr_delta≈+0.07 — on-target. UX call, not a tuning bug (§11)	Low
7	Investigate 1v1 side-A bias (~4pp at parity)	Initiative/turn-order edge in duels (§7)	Low
8	Keep finite max-ticks + add sudden-death	Bruiser/warrior stalls hit 24% timeout; mega2's 1e6 hid this (§10)	Low

Sequence: do #1 then re-sim. Don't trust hybrid/affinity verdicts until mage is fixed — a broken role drags every opponent's win rate and distorts the field. Re-run a full mega (incl. team3 snow+thunder) after the mage buff and re-read wr_delta.

13. What's trustworthy vs provisional

- **Solid (structural), confirmed at mega4's 5× sample:** aggregate faction balance, the cliff, team-size variance shrink, the win-curve shape, the mage collapse (controlled for tier **and** faction), the kit-flip direction. win_rate is **identical mega3↔mega4 in 1v1** (full round-robin, same battles); team2/team3 use freshly sampled matchups so they drift ≤0.002 — still well within convergence.
- **Now trustworthy (after the fix):** wr_delta as a per-piece tuning residual vs the fixed P target. The mega3 wr_delta numbers in §6/§11 were on the buggy metric — use the §0.5 corrected values.
- **Provisional:** absolute role/affinity numbers shift once the mage kit is buffed (mage opponents' win rates are inflated by farming mages). **team3 mega4 is incomplete (4/6 weathers)** — its wr_delta will firm up when snow+thunder land.
- **Caveat:** weather samples are sane but the +0.008–0.011 effect is small enough that any per-affinity number is within noise; treat §9 as directional.

14. Reproducibility

All scripts in reviews/mega_sim/, base R only (no external packages):

script	output
00_load.R	loader helpers (power, load_ratings(dir), load_results(dir), team_power); schema-tolerant across mega3/mega4
01_analysis.R	core mega3 stats → tables/*.csv, cache.rds
02_results.R	per-battle win-curve + mega2 comparison → cache_results.rds
03_plots.R	mega3 plots 01–10 → plots/
04_mega4.R	mega4 load + corrected-model stats + mega3↔mega4 comparison → cache_mega4.rds
05_mega4_plots.R	corrected-model plots 11–13 → plots/

```
cd /home/merlindk/development/tempest-fauna-trail
Rscript reviews/mega_sim/01_analysis.R # mega3 core
Rscript reviews/mega_sim/02_results.R # mega3 win-curve + mega2 cmp
Rscript reviews/mega_sim/03_plots.R # mega3 plots
Rscript reviews/mega_sim/04_mega4.R # mega4 + corrected expected_wr
Rscript reviews/mega_sim/05_mega4_plots.R # calibration-fix plots
```

Tables: tables/{role_balance,affinity_balance,tier_curve,weather_sensitivity,piece_overall,wincurve,mega2_vs_mega3_role,mega3_vs_mega4_role,mega4_piece_overall}.csv.

Note on mega4 schema: mega4 ratings dropped the Bradley-Terry columns (beta, beta_ratio, beta_deviation_pct) when expected_wr moved to the deterministic power-threshold model. 00_load.R handles both via load_ratings(dir, common=TRUE).